EARTHQUAKE RISK AND RESILIENT CONSTRUCTION

SEISMIC RISK GRAMMAR → MAGNITUDE-INTENSITY FIREWALL → ZONING-TO-SITE LADDER → SITE-EFFECT MAP → VULNERABILITY CHAIN → MITIGATION MATRIX

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW · ◆ SUBORDINATE EXTRA · ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

00

STAGE 00 SEISMIC RISK GRAMMAR

SEISMIC RISK GRAMMAR · GROUND SHAKING AND SECONDARY EFFECTS · PEOPLE BUILDINGS INFRASTRUCTURE LIFELINES · SITE + DESIGN + CONSTRUCTION + MAINTENANCE · PREPAREDNESS RESPONSE AND RECOVERY

HAZARD → ground shaking and secondary effects

EXPOSURE → people buildings infrastructure lifelines → VULNERABILITY → site + design + construction + maintenance → CAPACITY → preparedness response and recovery

HAZARD / RISK / RESILIENCE CONCEPT
- HAZARD → ground shaking and secondary effects
- EXPOSURE → people buildings infrastructure lifelines

INSTITUTIONAL OR COMMUNITY RESPONSE
- VULNERABILITY → site + design + construction + maintenance
- CAPACITY → preparedness response and recovery

CAPACITY / STATUS / EVIDENCE LIMIT
- HAZARD → ground shaking and secondary effects
- EXPOSURE → people buildings infrastructure lifelines

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VULNERABILITY → site + design + construction + maintenance.

01

STAGE 01 MAGNITUDE-INTENSITY FIREWALL

MAGNITUDE-INTENSITY FIREWALL · EVENT SIZE · ENERGY MEASURE · OBSERVED EFFECTS AT A LOCATION · ONE EVENT · VARYING LOCAL INTENSITIES · SCALES ARE NOT INTERCHANGEABLE

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
MAGNITUDE → event size / energy measure	ONE EVENT → varying local intensities	MAGNITUDE → event size / energy measure
INTENSITY → observed effects at a location	TRAP → scales are not interchangeable	INTENSITY → observed effects at a location

HAZARD / ACTOR / INSTRUMENT
- MAGNITUDE → event size / energy measure
- INTENSITY → observed effects at a location

DECISIVE RISK OR MANDATE DISTINCTION
- ONE EVENT → varying local intensities
- TRAP → scales are not interchangeable

WARNING / IMPLEMENTATION / OUTCOME LIMIT
- MAGNITUDE → event size / energy measure
- INTENSITY → observed effects at a location

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ONE EVENT → varying local intensities; TRAP → scales are not interchangeable.

02

STAGE 02 ZONING-TO-SITE LADDER

ZONING-TO-SITE LADDER · SEISMIC ZONE · BROAD EXPECTED HAZARD · FINER LOCAL VARIATION · SITE ASSESSMENT · PROJECT-SPECIFIC CONDITIONS · NONE OF THESE · PREDICTS THE NEXT EVENT

SEISMIC ZONE → broad expected hazard

MICROZONATION → finer local variation → SITE ASSESSMENT → project-specific conditions → NONE OF THESE → predicts the next event

RISK CONDITION / INPUT
- SEISMIC ZONE → broad expected hazard
- MICROZONATION → finer local variation

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM
- SITE ASSESSMENT → project-specific conditions
- NONE OF THESE → predicts the next event

LAST-MILE / INCLUSION / OUTCOME LIMIT
- SEISMIC ZONE → broad expected hazard
- MICROZONATION → finer local variation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NONE OF THESE → predicts the next event.

03

STAGE 03 SITE-EFFECT MAP

SITE-EFFECT MAP · SOIL CONDITION · MODIFIES SHAKING OR SECONDARY RISK · BUILT FORM + ACCESS · MODIFIES EXPOSURE RESPONSE AND LOSS

GROUND / SLOPE / SOIL CONDITION

modifies shaking or secondary risk → BUILT FORM + ACCESS → modifies exposure response and loss

RISK CONDITION / INPUT
- GROUND / SLOPE / SOIL CONDITION
- modifies shaking or secondary risk

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM
- BUILT FORM + ACCESS
- modifies exposure response and loss

LAST-MILE / INCLUSION / OUTCOME LIMIT
- GROUND / SLOPE / SOIL CONDITION
- modifies shaking or secondary risk

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BUILT FORM + ACCESS; modifies exposure response and loss.

04

STAGE 04 VULNERABILITY CHAIN

VULNERABILITY CHAIN · IRREGULAR FORM · WEAK CONNECTION · POOR WORKMANSHIP · UNAUTHORISED ALTERATION · DEGRADED MATERIAL · FAILED APPROVAL · COLLAPSE OR SERVICE DISRUPTION RISK

IRREGULAR FORM

WEAK CONNECTION → POOR WORKMANSHIP → UNAUTHORISED ALTERATION

RISK CONDITION / INPUT
- IRREGULAR FORM
- WEAK CONNECTION
- POOR WORKMANSHIP

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM
- UNAUTHORISED ALTERATION
- DEGRADED MATERIAL
- FAILED APPROVAL / INSPECTION / MAINTENANCE

LAST-MILE / INCLUSION / OUTCOME LIMIT
- RESULT → collapse or service disruption risk

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: IRREGULAR FORM | WEAK CONNECTION | POOR WORKMANSHIP.

05

STAGE 05 MITIGATION MATRIX

MITIGATION MATRIX · SAFE NEW BUILD + ASSESSED STRENGTHENING · LAND USE + ENFORCEMENT + DRILLS · SECURE FIXTURES EQUIPMENT UTILITIES · COMBINE; DO NOT SUBSTITUTE ONE FOR ALL

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
STRUCTURAL → safe new build + assessed strengthening	CONTENTS → secure fixtures equipment utilities	STRUCTURAL → safe new build + assessed strengthening
NON-STRUCTURAL → land use + enforcement + drills	RULE → combine; do not substitute one for all	NON-STRUCTURAL → land use + enforcement + drills

HAZARD / ACTOR / INSTRUMENT
- STRUCTURAL → safe new build + assessed strengthening
- NON-STRUCTURAL → land use + enforcement + drills

DECISIVE RISK OR MANDATE DISTINCTION
- CONTENTS → secure fixtures equipment utilities
- RULE → combine; do not substitute one for all

WARNING / IMPLEMENTATION / OUTCOME LIMIT
- STRUCTURAL → safe new build + assessed strengthening
- NON-STRUCTURAL → land use + enforcement + drills

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → combine; do not substitute one for all.

06

STAGE 06 CODE-TO-COMPLIANCE RAIL

CODE-TO-COMPLIANCE RAIL · 1 STANDARD · 2 COMPETENT DESIGN · 3 TRAINED CONSTRUCTION · 4 APPROVAL

1 STANDARD / BYE-LAW

2 COMPETENT DESIGN → 3 TRAINED CONSTRUCTION → 4 APPROVAL → INSPECTION → MAINTENANCE

HAZARD / RISK / RESILIENCE CONCEPT
- 1 STANDARD / BYE-LAW
- 2 COMPETENT DESIGN

INSTITUTIONAL OR COMMUNITY RESPONSE
- 3 TRAINED CONSTRUCTION
- 4 APPROVAL → INSPECTION → MAINTENANCE

CAPACITY / STATUS / EVIDENCE LIMIT
- 1 STANDARD / BYE-LAW
- 2 COMPETENT DESIGN

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 3 TRAINED CONSTRUCTION; 4 APPROVAL → INSPECTION → MAINTENANCE.

07

STAGE 07 DUCTILITY CONCEPT BOX

DUCTILITY CONCEPT BOX · DEFORMATION CAPACITY · CONTROLLED ENERGY DISSIPATION · SUPPORTS LIFE-SAFETY OBJECTIVE · CONCEPT ONLY, NO DETAILING INSTRUCTION

DEFORMATION CAPACITY

controlled energy dissipation → supports life-safety objective → BOUNDARY → concept only, no detailing instruction

HAZARD / RISK / RESILIENCE CONCEPT
- DEFORMATION CAPACITY
- controlled energy dissipation

INSTITUTIONAL OR COMMUNITY RESPONSE
- supports life-safety objective
- BOUNDARY → concept only, no detailing instruction

CAPACITY / STATUS / EVIDENCE LIMIT
- DEFORMATION CAPACITY
- controlled energy dissipation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: supports life-safety objective; BOUNDARY → concept only, no detailing instruction.

08

STAGE 08 RETROFIT PRIORITY FUNNEL

RETROFIT PRIORITY FUNNEL · SCREEN EXISTING STOCK · PRIORITISE LIFELINES + HIGH OCCUPANCY + HIGH VULNERABILITY · DETAILED ASSESSMENT · LISTING ALONE · NOT COMPLETION

SCREEN EXISTING STOCK

PRIORITISE LIFELINES + HIGH OCCUPANCY + HIGH VULNERABILITY → DETAILED ASSESSMENT → SANCTION → WORK → VERIFY → AUDIT / LISTING ALONE → NOT COMPLETION

HAZARD / RISK / RESILIENCE CONCEPT
- SCREEN EXISTING STOCK
- PRIORITISE LIFELINES + HIGH OCCUPANCY + HIGH VULNERABILITY

INSTITUTIONAL OR COMMUNITY RESPONSE
- DETAILED ASSESSMENT → SANCTION → WORK → VERIFY
- AUDIT / LISTING ALONE → NOT COMPLETION

CAPACITY / STATUS / EVIDENCE LIMIT
- SCREEN EXISTING STOCK
- PRIORITISE LIFELINES + HIGH OCCUPANCY + HIGH VULNERABILITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DETAILED ASSESSMENT → SANCTION → WORK → VERIFY.

09

STAGE 09 LIFELINE CONTINUITY MAP

LIFELINE CONTINUITY MAP · HOSPITAL + EOC + FIRE · POWER + WATER + TELECOM · PHYSICAL SAFETY AND POST-EVENT FUNCTION

HOSPITAL + EOC + FIRE / RESCUE

POWER + WATER + TELECOM → ROAD / BRIDGE / ACCESS → TEST → physical safety AND post-event function

HAZARD / RISK / RESILIENCE CONCEPT
- HOSPITAL + EOC + FIRE / RESCUE
- POWER + WATER + TELECOM

INSTITUTIONAL OR COMMUNITY RESPONSE
- ROAD / BRIDGE / ACCESS
- TEST → physical safety AND post-event function

CAPACITY / STATUS / EVIDENCE LIMIT
- HOSPITAL + EOC + FIRE / RESCUE
- POWER + WATER + TELECOM

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ROAD / BRIDGE / ACCESS; TEST → physical safety AND post-event function.

10

STAGE 10 MONITORING-PREDICTION BOUNDARY

MONITORING-PREDICTION BOUNDARY · EVENT DETECTION · PARAMETER REPORT · HAZARD ASSESSMENT · PLANNING INPUT · PROTECTIVE ACTION · PRECISE FUTURE MAGNITUDE PLACE AND TIME

SENSOR → EVENT DETECTION → PARAMETER REPORT → HAZARD ASSESSMENT → PLANNING INPUT → PREPAREDNESS → PROTECTIVE ACTION → NOT → precise future magnitude place and time

HAZARD / ACTOR / INSTRUMENT
- SENSOR → EVENT DETECTION → PARAMETER REPORT
- HAZARD ASSESSMENT → PLANNING INPUT

DECISIVE RISK OR MANDATE DISTINCTION
- PREPAREDNESS → PROTECTIVE ACTION
- NOT → precise future magnitude place and time

WARNING / IMPLEMENTATION / OUTCOME LIMIT
- SENSOR → EVENT DETECTION → PARAMETER REPORT
- HAZARD ASSESSMENT → PLANNING INPUT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NOT → precise future magnitude place and time.

11

STAGE 11 EARTHQUAKE ANSWER SPINE

EARTHQUAKE ANSWER SPINE · DEFINE RISK · SEPARATE MAGNITUDE INTENSITY · MAP ZONE + SITE + EXPOSURE + VULNERABILITY · COMBINE CODE RETROFIT LAND USE CONTENTS AND LIFELINES · TEST ENFORCEMENT EQUITY CONTINUITY AND VERIFIED OUTCOME

DEFINE RISK → SEPARATE MAGNITUDE INTENSITY → MAP ZONE + SITE + EXPOSURE + VULNERABILITY → COMBINE CODE RETROFIT LAND USE CONTENTS AND LIFELINES → TEST ENFORCEMENT EQUITY CONTINUITY AND VERIFIED OUTCOME

HAZARD / RISK / RESILIENCE CONCEPT
- DEFINE RISK → SEPARATE MAGNITUDE INTENSITY
- MAP ZONE + SITE + EXPOSURE + VULNERABILITY

INSTITUTIONAL OR COMMUNITY RESPONSE
- COMBINE CODE RETROFIT LAND USE CONTENTS AND LIFELINES
- TEST ENFORCEMENT EQUITY CONTINUITY AND VERIFIED OUTCOME

CAPACITY / STATUS / EVIDENCE LIMIT
- DEFINE RISK → SEPARATE MAGNITUDE INTENSITY
- MAP ZONE + SITE + EXPOSURE + VULNERABILITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COMBINE CODE RETROFIT LAND USE CONTENTS AND LIFELINES.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · RESPONSIBILITY IS EXPLICITLY BIFURCATED BY STRUCTURE TYPE · REGULATION AND ENFORCEMENT MECHANISM · STATE GOVERNMENTS MUST · LIFELINE-STRUCTURE IDENTIFICATION IS STATE-SPECIFIC, NOT · HIMALAYAN-BELT DEVELOPMENT PRESSURE · FOURTEEN STATES LIE IN THE · EARTHQUAKE EARLY WARNING IS NOT THE SAME AS EARTHQUAKE

OPTIONAL RISK / RESILIENCE DEPTH
- Responsibility is explicitly bifurcated by structure type : all
- Regulation and enforcement mechanism : State Governments must
- Lifeline-structure identification is State-specific, not
- CAUTION: Himalayan-belt development pressure : fourteen States lie in the

MANDATE, WARNING OR OUTCOME LIMIT
- NOT: Earthquake early warning is not the same as earthquake
- Any claim about current seismic-monitoring-station density, network
- CAUTION: The withdrawn seismic-zonation revision is the sharpest available
- CAUTION: Informal/rural construction bears disproportionate seismic risk :

COMPARATIVE HAZARD / GOVERNANCE NUANCE
- CAUTION: New-construction codes versus existing-stock retrofitting is a
- NOT: The "a National Building Code exists, therefore buildings are
- CAUTION: State-level variation in retrofitting pace (Section 3) is a
- CAUTION: VisionAS does not detail a specific seismic-retrofitting financing

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels